Chapter 3. Stratified Random Sampling

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1 Introduction

Definition 3.1

A **stratified random sample** is obtained by separating the population into non-overlapping groups called **strata** and then selecting a simple random sample from each stratum, independently across strata.

Because the strata are sampled independently, each stratum can be treated as a separate population. Stratification pays off when the variation *between* strata is large (the strata are internally homogeneous). Model-wise, SRS uses $y_i = \mu + \varepsilon_i$, while stratified sampling uses $y_{ij} = \mu_j + \varepsilon_{ij}$: each stratum has its own mean. Once the selection procedure is fixed, the design questions are

- how many strata to construct,
- how to allocate the total sample size to the strata,
- how to choose the strata.

2 Notation

Strata are indexed by $j = 1, \ldots, L$ and units within a stratum by i; y_{ij} is the observation on unit i of stratum j. Sampling within each stratum is SRS without replacement.

N_j, n_j	population and sample size of stratum j; $N = \sum N_j$, $n = \sum n_j$
$w_j = N_j/N$	stratum weight
$f_j = n_j/N_j$	sampling fraction in stratum j
$\mu_j = \frac{1}{N_j}\sum_{i=1}^{N_j} Y_{ij}$	true stratum mean
$\bar{y}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} y_{ij}$	sample stratum mean
$\sigma_j^2 = \frac{1}{N_j}\sum_i (Y_{ij}-\mu_j)^2$	true stratum variance
$S_j^2 = \frac{N_j}{N_j-1}\sigma_j^2$	stratum variance with divisor $N_j - 1$
$s_j^2 = \frac{1}{n_j-1}\sum_{i=1}^{n_j}(y_{ij}-\bar{y}_j)^2$	sample stratum variance

3 Estimation of the Population Mean

Motivation. If the population splits into groups that differ systematically, sampling each group separately removes the between-group variation from the error. The stratified estimator weights the stratum means by stratum sizes and is never worse than SRS under proportional allocation.

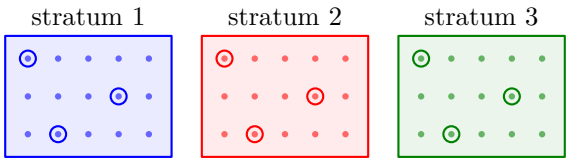


Figure. *Stratified random sampling: the population is split into homogeneous strata and an independent simple random sample (circled units) is drawn within each. Only within-stratum variation enters the variance of* $\bar{y}_{st} = \sum_h W_h \bar{y}_h$.

Theorem 3.2: Stratified estimator of μ

$$\bar{y}_{\text{st}} = \frac{1}{N}\sum_{j=1}^{L} N_j \bar{y}_j = \sum_{j=1}^{L} w_j \bar{y}_j$$

is unbiased for μ, with

$$\operatorname{Var}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\operatorname{Var}}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{s_j^2}{n_j}, \qquad \mathrm{E}[\widehat{\operatorname{Var}}(\bar{y}_{\text{st}})] = \operatorname{Var}(\bar{y}_{\text{st}}).$$

A $100(1-\alpha)\%$ interval is $\bar{y}_{\text{st}} \pm z_{\alpha/2}\sqrt{\widehat{\operatorname{Var}}(\bar{y}_{\text{st}})}$.

Proof. Each $\bar{y}_j$ is an SRS mean, so by Ch. 2, $\mathrm{E}(\bar{y}_j) = \mu_j$, $\operatorname{Var}(\bar{y}_j) = (1-f_j)S_j^2/n_j$ and $\mathrm{E}(s_j^2) = S_j^2$. Thus

$$\mathrm{E}(\bar{y}_{\text{st}}) = \sum_j w_j \mu_j = \frac{1}{N}(N_1\mu_1 + \cdots + N_L\mu_L) = \frac{1}{N}(\tau_1 + \cdots + \tau_L) = \frac{\tau}{N} = \mu.$$

Since the strata are sampled independently the covariances vanish and $\operatorname{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \operatorname{Var}(\bar{y}_j)$. Unbiasedness of $\widehat{\operatorname{Var}}$ follows from $\mathrm{E}(s_j^2) = S_j^2$. ∎

4 Estimation of the Population Total

Motivation. The stratified total is the sum of stratum totals; its variance adds over strata because samples in different strata are independent.

Theorem 3.3

$\hat{\tau}_{\text{st}} = N\bar{y}_{\text{st}} = \sum_{j=1}^{L} N_j \bar{y}_j$ is unbiased for τ, and

$$\text{Var}(\hat{\tau}_{\text{st}}) = N^2 \text{Var}(\bar{y}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1 - f_j) \frac{S_j^2}{n_j}, \qquad \widehat{\text{Var}}(\hat{\tau}_{\text{st}}) = \sum_{j=1}^{L} N_j^2 (1 - f_j) \frac{s_j^2}{n_j}.$$

Example 3.4: TV-viewing time

	N_j	w_j	n_j	$\bar{y}_j$	s_j
Town A	155	0.5	20	33.90	5.95
Town B	62	0.2	8	25.12	15.25
Rural	93	0.3	12	19.00	9.36
Total	310		40		

Mean.

$$\bar{y}_{\text{st}} = \frac{155(33.90) + 62(25.12) + 93(19.00)}{310} = \frac{8578.94}{310} = 27.67.$$

$$\begin{aligned}\widehat{\text{Var}}(\bar{y}_{\text{st}}) &= (0.5)^2 \frac{5.95^2}{20}\left(1 - \frac{20}{155}\right) + (0.2)^2 \frac{15.25^2}{8}\left(1 - \frac{8}{62}\right) + (0.3)^2 \frac{9.36^2}{12}\left(1 - \frac{12}{93}\right) \\ &= 0.385 + 1.013 + 0.572 = 1.97,\end{aligned}$$

so the estimate with a 2-SD bound is $27.67 \pm 2\sqrt{1.97} = 27.67 \pm 2.81$.
Total. $\hat{\tau}_{\text{st}} = 310(27.67) = 8579$, $\widehat{\text{Var}}(\hat{\tau}_{\text{st}}) = 310^2(1.97) \approx 189{,}400$, and the bound is $2\sqrt{189{,}400} \approx 870$.

5 Sample Size for Estimating μ and τ

Setting $2\sqrt{\text{Var}(\bar{y}_{\text{st}})} = B$ alone cannot be solved for n; we must also fix how n is split among the strata. Write the allocation as $n_j = n a_j$ with $\sum a_j = 1$.

Theorem 3.5: Sample size for a given allocation

$$n = \frac{\sum_{j=1}^{L} N_j^2 S_j^2 / a_j}{N^2 D + \sum_{j=1}^{L} N_j S_j^2}, \qquad D = \begin{cases} B^2/4 & \text{for } \mu, \\ B^2/(4N^2) & \text{for } \tau, \end{cases}$$

with S_j^2 replaced by prior estimates s_j^2 in practice.

Derivation

For μ let $D = (B/2)^2 = \text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}(1 - \frac{n_j}{N_j})$. Multiply by N^2 and substitute $n_j = n a_j$:

$$N^2 D = \sum_j N_j^2 \frac{S_j^2}{n a_j} - \sum_j N_j S_j^2 \implies n\Big(N^2 D + \sum_j N_j S_j^2\Big) = \sum_j \frac{N_j^2 S_j^2}{a_j}.$$

For τ, $\text{Var}(\hat{\tau}_{\text{st}}) = N^2 \text{Var}(\bar{y}_{\text{st}}) = B^2/4$ gives the same formula with $D = B^2/(4N^2)$. (Using 1.96 instead of 2 replaces $D = (B/2)^2$ by $(B/1.96)^2$.)

Example 3.6: Equal allocation

For the TV population $(N_j = 155, 62, 93)$ with prior variances $s_1^2 = 25$, $s_2^2 = 225$, $s_3^2 = 100$, equal allocation $a_j = 1/3$ and $B = 2$ for μ $(D = 1)$:

$$n = \frac{3(155^2 \cdot 25 + 62^2 \cdot 225 + 93^2 \cdot 100)}{310^2(1) + (155 \cdot 25 + 62 \cdot 225 + 93 \cdot 100)} = \frac{6{,}991{,}275}{123{,}225} = 56.7,$$

so $n = 57$ and $n_1 = n_2 = n_3 = 19$.

6 Allocation of the Sample

Let the survey cost be

$$TC = c_0 + \sum_{j=1}^{L} c_j n_j,$$

where c_0 is the overhead cost and c_j the cost per observation in stratum j.

6.1 Equal allocation

$n_j = n/L$. Then $TC = c_0 + \frac{n}{L}\sum c_j$, so $n = \dfrac{(TC - c_0)L}{\sum_j c_j}$.

6.2 Proportional allocation

$n_j = nw_j = nN_j/N$, i.e. $a_j = w_j$ and all $f_j = f$ are equal. Then $TC = c_0 + n\sum c_j w_j$, so $n = \dfrac{TC - c_0}{\sum_j c_j w_j}$; if $c_j \equiv c$, $n = (TC - c_0)/c$. The estimator becomes self-weighting: $\bar{y}_{\text{st}}$ equals the plain mean of all n observations.

6.3 Optimum allocation

Theorem 3.7: Optimum allocation

For a fixed total cost (equivalently, for a fixed variance) $\text{Var}(\bar{y}_{\text{st}})$ is minimized when

$$n_j \propto \frac{w_j S_j}{\sqrt{c_j}}, \qquad \text{i.e.} \qquad n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}},$$

and the total sample size under a budget TC is

$$n = \frac{(TC - c_0)\sum_k w_k S_k/\sqrt{c_k}}{\sum_k w_k S_k \sqrt{c_k}}.$$

Derivation: Cauchy–Schwarz argument

Split the variance into the part depending on the allocation and a constant:

$$\text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\left(1 - \frac{n_j}{N_j}\right) = \underbrace{\sum_j \frac{w_j^2 S_j^2}{n_j}}_{V'} - \underbrace{\sum_j \frac{w_j S_j^2}{N}}_{\text{const}}.$$

Let $C' = TC - c_0 = \sum_j c_j n_j$. Minimizing V' for fixed C', or C' for fixed V', is the same as minimizing the product

$$V'C' = \Big(\sum_j \frac{w_j^2 S_j^2}{n_j}\Big)\Big(\sum_j c_j n_j\Big) = \Big(\sum_j a_j^2\Big)\Big(\sum_j b_j^2\Big), \quad a_j = \frac{w_j S_j}{\sqrt{n_j}},\ b_j = \sqrt{c_j n_j}.$$

By the Cauchy–Schwarz inequality

$$V'C' \geq \Big(\sum_j a_j b_j\Big)^2 = \Big(\sum_j w_j S_j \sqrt{c_j}\Big)^2,$$

whose right-hand side does not depend on the n_j. Equality holds iff b_j/a_j is constant:

$$\frac{b_j}{a_j} = \frac{n_j \sqrt{c_j}}{w_j S_j} = \text{const} \iff n_j \propto \frac{w_j S_j}{\sqrt{c_j}}.$$

Substituting $n_j = n \frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}}$ into $TC = c_0 + \sum c_j n_j$ gives $TC = c_0 + n\alpha$ with $\alpha = \frac{\sum_k w_k S_k \sqrt{c_k}}{\sum_k w_k S_k/\sqrt{c_k}}$, so $n = (TC - c_0)/\alpha$.

We therefore take a larger sample in a stratum that is (1) larger, (2) more variable, or (3) cheaper to sample.

6.4 Neyman allocation

Theorem 3.8: Neyman allocation

If the costs are equal, $c_j \equiv c$, the optimum allocation becomes

$$n_j = n \frac{w_j S_j}{\sum_k w_k S_k} = n \frac{N_j S_j}{\sum_k N_k S_k}.$$

Example 3.9: Neyman allocation for the TV data

With $n = 40$ and $N_j s_j = 922.3,\ 945.5,\ 870.5$ (sum 2738.2), the Neyman allocation is

$$n_1 = 13.5, \quad n_2 = 13.8, \quad n_3 = 12.7 \ \Rightarrow\ (13, 14, 13),$$

compared with the proportional allocation $(20, 8, 12)$ actually used. The small but very variable Town B gets many more observations. Treating s_j as S_j, the variance of $\bar{y}_{\text{st}}$ drops from 1.97 (proportional) to about 1.66 (Neyman).

7 Estimation of a Population Proportion

Motivation. A proportion is the mean of a 0–1 variable, so the stratified formulas apply directly with $p_h(1 - p_h)$ in place of the stratum variances.

With $y_{ij} \in \{0, 1\}$ the stratum mean $\bar{y}_j$ is the stratum proportion $\hat{p}_j$, and

$$\hat{p}_{\text{st}} = \frac{1}{N}(N_1 \hat{p}_1 + \cdots + N_L \hat{p}_L) = \sum_j w_j \hat{p}_j, \qquad \mathrm{E}(\hat{p}_{\text{st}}) = \sum_j w_j p_j = p.$$

Theorem 3.10

$$\mathrm{Var}(\hat{p}_{\mathrm{st}}) = \sum_j w_j^2 \frac{p_j q_j}{n_j}\Big(\frac{N_j - n_j}{N_j - 1}\Big), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\mathrm{st}}) = \sum_j w_j^2 \frac{\hat{p}_j \hat{q}_j}{n_j - 1}(1 - f_j),$$

and a $100(1-\alpha)\%$ interval for p is $\hat{p}_{\mathrm{st}} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{p}_{\mathrm{st}})}$.

These follow from Ch. 2 applied stratum by stratum, the divisor $n_j - 1$ making each term unbiased.

8 Selected exercises

Motivation. The allocation theorems say which allocation is best but not by how much. The first two items give exact expressions for the gains of Neyman over proportional allocation and of proportional allocation over SRS; the third is a full design computation with unequal costs.

Exercise 3.1 (Proportional versus Neyman allocation)

Let the costs be equal and the total sample size n fixed. Denote by $\bar{y}_{\mathrm{prop}}$ and $\bar{y}_{\mathrm{opt}}$ the stratified means under proportional allocation $n_j = nw_j$ and under Neyman allocation $n_j = nw_jS_j/\bar{S}$, where $\bar{S} = \sum_j w_j S_j$ (assume $n_j \le N_j$). Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\sum_{j=1}^{L} w_j (S_j - \bar{S})^2,$$

so $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) > \mathrm{Var}(\bar{y}_{\mathrm{opt}})$ unless all S_j are equal.

Solution.

Step 1. (A common constant.) For any allocation,

$$\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\Big(1 - \frac{n_j}{N_j}\Big) = \sum_j \frac{w_j^2 S_j^2}{n_j} - \frac{1}{N}\sum_j w_j S_j^2,$$

since $w_j^2/N_j = w_j/N$. The last sum does not depend on the allocation.

Step 2. (Proportional.) With $n_j = nw_j$, $\sum_j w_j^2 S_j^2/n_j = \frac{1}{n}\sum_j w_j S_j^2$.

Step 3. (Neyman.) With $n_j = nw_jS_j/\bar{S}$, $\sum_j w_j^2 S_j^2/n_j = \frac{\bar{S}}{n}\sum_j w_j S_j = \frac{\bar{S}^2}{n}$.

Step 4. (Difference.) Subtracting, the constants of Step 1 cancel and

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\Big(\sum_j w_j S_j^2 - \bar{S}^2\Big) = \frac{1}{n}\sum_j w_j (S_j - \bar{S})^2,$$

the last step being the identity $\sum_j w_j(S_j - \bar{S})^2 = \sum_j w_j S_j^2 - 2\bar{S}\sum_j w_j S_j + \bar{S}^2 \sum_j w_j = \sum_j w_j S_j^2 - \bar{S}^2$ (recall $\sum_j w_j = 1$).

Step 5. The right side is a weighted variance of the stratum standard deviations, with positive weights; it vanishes iff $S_j = \bar{S}$ for all j, in which case the two allocations coincide. The gain from Neyman allocation is therefore large exactly when the strata differ in variability. ■

Exercise 3.2 (Proportional allocation versus SRS)

Let $\bar{y}_{\text{ran}}$ be the mean of an SRS of size n from the whole population, with variance $\frac{1-f}{n}S^2$. Show that

$$\operatorname{Var}(\bar{y}_{\text{ran}}) - \operatorname{Var}(\bar{y}_{\text{prop}}) = \frac{1-f}{n(N-1)}\Big[\sum_{j=1}^{L} N_j(\mu_j-\mu)^2 - \sum_{j=1}^{L}(1-w_j)S_j^2\Big],$$

and conclude that proportional allocation is better than SRS unless the stratum means are nearly equal.

Solution. [Supplement]

Step 1. (Analysis of variance of the population.) Adding and subtracting μ_j inside stratum j, the cross term vanishes because $\sum_i (Y_{ij}-\mu_j)=0$:

$$(N-1)S^2 = \sum_j\sum_i (Y_{ij}-\mu)^2 = \sum_j (N_j-1)S_j^2 + \sum_j N_j(\mu_j-\mu)^2.$$

Step 2. Under proportional allocation $f_j = f$, so $\operatorname{Var}(\bar{y}_{\text{prop}}) = \sum_j w_j^2(1-f)\frac{S_j^2}{nw_j} = \frac{1-f}{n}\sum_j w_j S_j^2$. Hence

$$\operatorname{Var}(\bar{y}_{\text{ran}}) - \operatorname{Var}(\bar{y}_{\text{prop}}) = \frac{1-f}{n}\Big(S^2 - \sum_j w_j S_j^2\Big).$$

Step 3. By Step 1,

$$S^2 - \sum_j w_j S_j^2 = \frac{\sum_j N_j(\mu_j-\mu)^2}{N-1} + \sum_j S_j^2\Big(\frac{N_j-1}{N-1} - \frac{N_j}{N}\Big),$$

and $\frac{N_j-1}{N-1} - \frac{N_j}{N} = \frac{N(N_j-1)-N_j(N-1)}{N(N-1)} = -\frac{N-N_j}{N(N-1)} = -\frac{1-w_j}{N-1}$. Substituting gives the claim.

Step 4. The first sum in the bracket is the between-strata sum of squares, which grows with the stratum sizes N_j; the second is a sum of L within-stratum variances with coefficients below 1. Unless the μ_j are nearly equal, the first dominates and $\operatorname{Var}(\bar{y}_{\text{prop}}) < \operatorname{Var}(\bar{y}_{\text{ran}})$; with large N_j the difference is essentially $\frac{1-f}{n}\sum_j w_j(\mu_j-\mu)^2 \geq 0$. ■

Exercise 3.3 (Allocation with unequal costs)

A county plans to estimate the proportion of families who will use an expanded day-care facility. Families are stratified into users in the city, users in the country, nonusers in the city and nonusers in the country, with $N_j = 97, 43, 145, 68$. About 90% of present users and 50% of present nonusers are expected to use the expanded facility. An observation costs 4 for a user and 8 for a nonuser. Find the approximate sample size and allocation needed to estimate the proportion with a bound $B = 0.05$.

Solution. For 0/1 data $S_j^2 = \frac{N_j}{N_j-1}p_jq_j \approx p_jq_j$, so we use $p_jq_j = 0.09, 0.09, 0.25, 0.25$ in place of S_j^2.

Step 1. (Allocation.) By the optimum-allocation theorem, $n_j \propto w_jS_j/\sqrt{c_j} \propto N_j\sqrt{p_jq_j/c_j}$:

$$N_j\sqrt{p_jq_j/c_j} = 14.55,\ 6.45,\ 25.63,\ 12.02, \qquad \text{sum } 58.65,$$

so $a_j = 0.248,\ 0.110,\ 0.437,\ 0.205$.

Step 2. (Sample size.) With $N = 353$, $D = B^2/4 = 0.000625$ and the sample-size theorem,

$$n = \frac{\sum_j N_j^2 p_j q_j / a_j}{N^2 D + \sum_j N_j p_j q_j} = \frac{22{,}595}{77.88 + 65.85} = \frac{22{,}595}{143.73} = 157.2,$$

where $\sum_j N_j^2 p_j q_j / a_j = 846.8/0.248 + 166.4/0.110 + 5256.3/0.437 + 1156.0/0.205$ and $\sum_j N_j p_j q_j = 8.73 + 3.87 + 36.25 + 17.00$.

Step 3. Take $n = 158$; then $n_j = n a_j = 39.2,\ 17.4,\ 69.0,\ 32.4$, i.e. $(n_1, n_2, n_3, n_4) = (39, 17, 69, 33)$. Nonusers get more observations despite the higher cost because their proportion is more variable ($pq = 0.25$ versus 0.09); the cost factor $1/\sqrt{c_j}$ only partly offsets this. ■